A Novel Framework for Solar Panel Segmentation From Remote Sensing Images: Utilizing Chebyshev Transformer and Hyperspectral Decomposition

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***Abstract*— Solar panel segmentation (SPS) is identifying and locating solar panels from remote sensing images, such as aerial or satellite imagery. SPS is critical for energy monitoring, urban planning, and environmental studies, as it can provide information on the distribution and deployment of solar energy systems and their impact on the climate and the economy. However, the existing methods face several challenges, such as low-quality remote sensing images, varying resolutions, and high computational costs. These factors make it challenging to distinguish solar panels from other objects or backgrounds and accurately and efficiently segment them. This article proposes a novel hyperspectral solar segmentation network (HSS-Net) method for SPS, combining Chebyshev transformation (CHT) and hyperspectral synthetic decomposition (HSD). Our method can enhance the image quality, select the optimal bands, and segment the solar panels. We validated the presented method on three publicly available SPS benchmark datasets, such as base de données apprentissage profond PV (BDAPPV), photovoltaic (PV), and DeepSolar. We compared the performance of HSS-Net with the state-of-the-art (SOTA) methods, including convolutional neural network (CNN)-based and transformer-based networks and existing hyperspectral segmentation techniques. We used the intersection over union (IoU), the *F*1-score, and the Kappa coefficient (KI) as the evaluation metrics. We demonstrated that HSS-Net significantly surpasses SOTA methods in terms of accuracy, efficiency, and scalability, can advance remote sensing applications, and may provide more precise results in various relevant fields.**

***Index Terms*— Band selection (BS), hyperspectral imaging (HSI), remote sensing segmentation, solar panels segmentation.**

I. Introduction

SOLAR ENERGY, a renewable and clean power source, plays a vital role in reducing greenhouse gas emissions and fossil fuel dependence [1]. Effective management and monitoring of solar energy systems, specifically solar panel segmentation (SPS) in aerial images, is essential for optimal performance. This segmentation provides valuable insights for solar potential estimation, defect detection, environmental impact assessment, and solar system installation and expansion planning. However, SPS faces significant challenges due to the complexity and diversity of aerial images. Common issues include varying image resolutions, solar panels blending with surroundings due to similar textures (like pools or tennis courts), color inconsistencies, and different ground sampling distances (GSDs) in images. These challenges hinder the generalizability of deep learning-based segmentation methods, leading to a gap in fair method comparison, stated by Li et al. [2]. Many studies use private databases, which limit comparative analysis.

Recent advancements in fully convolutional networks (FCNs) [9] have improved the performance of image segmentation tasks. Various models have been proposed recently, such as U-Net-based models, the DeepLab model family [11], and PSP-Net [12]. Several studies have applied these architectures for SPS. Li et al. [13] proposed a cross-learning-based U-Net architecture to segment rooftop photovoltaic (PV) panels from satellite images. Yuan et al. [14] performed large-scale SPS with deep convolutional networks and achieved PV segmentation using only ∼3000 low-resolution Google satellite images [15]. These methods have achieved higher accuracy than older ones. However, they focused on general machine learning segmentation frameworks and did not consider the characteristics of the task. The main key aspects of SPS are the type of images used, typically remote sensing or aerial images, GSD variations, weather conditions, and roof conditions, which require a specific approach to have a generalized model. de Luis et al. [16] and Guo et al. [17] tried to utilize recent vision transformers to capture more global context, which can be beneficial for distinguishing solar panels from similarly colored or shaped objects in a wide area. Visual transformers (ViTs) can also handle different image resolutions efficiently, which is beneficial for aerial images. The drawbacks of using vision transformers include the need for significant computational power and a large volume of training data. These factors are critical in SPS, where aerial imaging may pose hardware constraints for real-time performance on limited devices. Moreover, having annotated aerial training datasets is costly compared to other segmentation tasks.

In recent years, Zhu et al. [46] proposed a deep solar PV refiner by incorporating an attention mechanism with DeepLabv3+ and PointRed module to handle small-size SPS from aerial images. Li et al. [47] analyzed the structural

Manuscript received 2 February 2024; revised 8 March 2024; accepted 5 April 2024. Date of publication 8 April 2024; date of current version 19 April 2024. This work was supported by the Advance Research Grants provided by the Foundation for Armenian Science and Technology, funded by Sarkis and Nune Sepetjians. *(Corresponding author: Hayk A. Gasparyan.)*

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Digital Object Identifier 10.1109/TGRS.2024.3386402

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factors of deep learning-based methods for effective feature extraction of PV array features. In addition to segmentation-based methods, the detector DetEEktor was employed by Schulz et al. [48] to detect PV panels and other types of renewable energy systems from aerial images. These studies have highlighted the importance of accuracy, inference speed, and algorithm efficiency. However, they have yet to explicitly solve the challenges present in data, such as data imbalance, skewed distributions, and different resolution sizes of aerial images [49].

A robust and adaptive framework is essential to tackle the unique demands of SPS effectively. This framework should be versatile enough to handle various use cases, regions, GSDs, and resolutions. A promising approach is incorporating physical attributes, such as color and texture, which remain consistent across different regions and resolutions. These attributes are often captured in hyperspectral images (HSIs), consisting of multiple bands containing distinct information. Research [19] indicates that using standard statistical target detection methods, solar panels are detectable in HSI. These methods include the adaptive cosine estimator, and the spectral verification process helps to exclude pixels lacking the key spectral characteristics of solar panel reflectance.

Jörges et al. [18] employed spaceborne HSI data from the PRISMA satellite at a 30-m resolution, achieving results comparable to human-level accuracy. However, the accessibility of HSI from satellites is limited, and employing hyperspectral cameras is often expensive. To address this, recent advancements focus on synthesizing HSIs from standard RGB images. A key example is the NTIRE 2022 challenge, which concentrated on spectral reconstruction from RGB images using 1000 training pairs of RGB images and their corresponding hyperspectral cubes. A comprehensive review [20] details the performances and results of the participants in this challenge.

Overall, current techniques often fail to adequately consider the intrinsic physical characteristics of solar panels, such as color and texture. They also commonly depend on low-resolution aerial images, which poses a challenge in differentiating small panels from their surroundings, especially at varying resolutions. Moreover, their high computational demands limit real-time applications.

This article introduces a novel framework that integrates hyperspectral decomposition (HSD) and Chebyshev transformations (CHTs) for a more accurate and efficient SPS. This approach marks the first use of spectral reconstruction techniques for SPS tasks. The key contributions of our framework include the following.

1) Enhanced HSD and CHT-based segmentation, improving accuracy while reducing computational time.
2) An unsupervised enhancement module for preprocessing low-quality images, featuring custom loss functions to optimize HSD for better spectral decomposition.
3) A trainable band selection (BS) module, selecting and combining bands to optimize computation and minimize redundant data.
4) CHT layers, reducing the number of trainable parameters and innovating the segmentation process.
5) *Extensive Comparative Analysis:* Our method undergoes detailed comparisons with 14 advanced segmentation techniques, particularly for aerial imagery, using public solar panel datasets.

The structure of this article is as follows. Section II describes the proposed method and its modules, Section III reports the experimental results and ablation studies, and Section IV concludes this article and suggests avenues for future work.

II. Methodology

In this section, we present the pipeline of HSS-Net (see Fig. 1). Initially, the input image is passed into the unsupervised enhancement module to highlight colors more distinctly. Following this, an HSD is used to distribute the image's information across various spectral bands. A subsequent BS module is then employed to generate weight maps, which are used for the weighted sum combination of these bands into a single guide image. The resulting guide image, along with the enhanced input image, is then fed into the CHT-based final segmentation network, which is responsible for predicting the output mask. Further details of each component within the model are discussed in the subsequent chapters.

A. Unsupervised Enhancement Module

The unsupervised enhancement module preprocesses low-quality aerial images and improves their contrast and features. The module has two submodules: a contrast enhancement submodule [43] and a feature enhancement submodule. The contrast enhancement submodule uses histogram equalization to adjust the contrast of the input image. This step is important for detection or segmentation tasks, where objects are small or indistinct, such as SPS. Our experiments show that HSD is more effective when the input RGB image has clear colors and high variance in the color histogram. We designed a lightweight enhancement module with about 15 K trainable parameters, inspired by Li et al. [23]. The module uses color curve estimation layers and a quadratic curve to refine the image's colors. The module uses a small neural network to generate 24 alpha maps, which are applied iteratively to the RGB channels

$$\mathrm{CE}_n(\mathbf{x}) = \mathrm{CE}_{n-1}(\mathbf{x}) + A_n(\mathbf{x})\mathrm{CE}_{n-1}(\mathbf{x}) - A_n(\mathbf{x})\mathrm{CE}_{n-1}(\mathbf{x})^2_{n-1}$$

where $A_n(\mathbf{x})$ (called the alpha map) is derived from a small neural network. This network consists of five convolutional layers, each with 32 kernels of size 3×3, followed by ReLU activation functions. Overall, 24 alpha maps are predicted, spanning eight iterations across each of the RGB channels. After iteratively applying a quadratic transformation, the network outputs an enhanced image with clear and distinguishable colors and high contrast [see Fig. 2(a)]. This enhancement is driven by custom-designed unsupervised loss functions, enabling the module to learn without the need for image-label pairs (see Section II-E). We provide the examples of the enhancement results in the simulation section. Notably, the advantage of this module is not only the effective image

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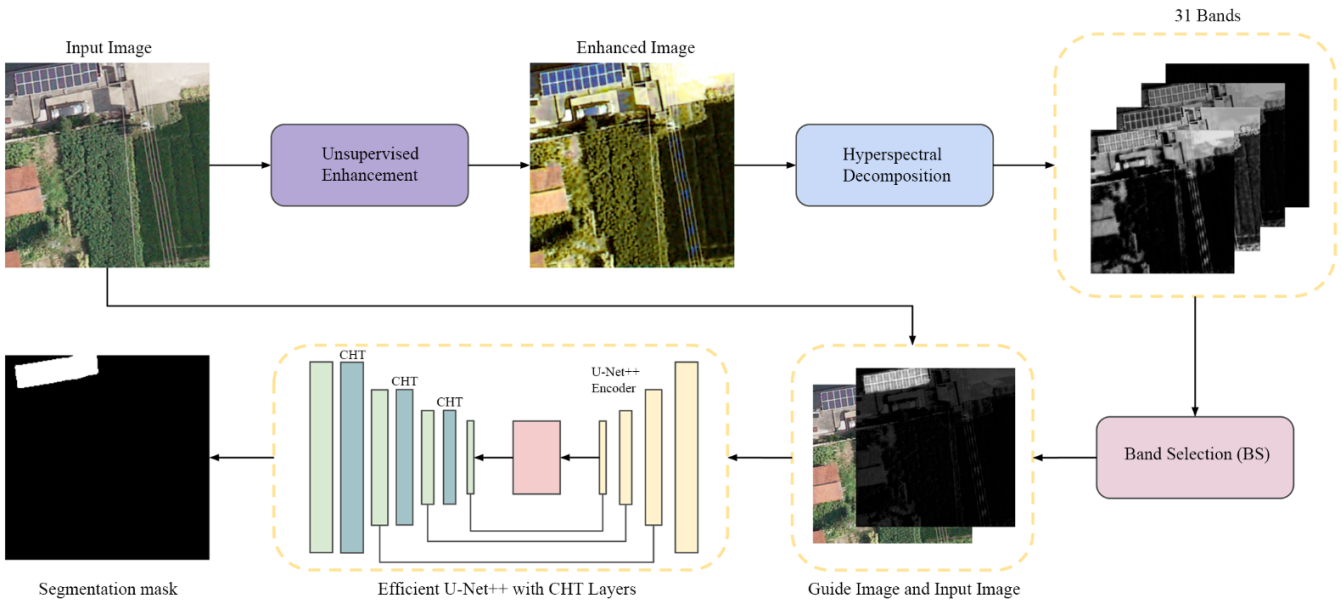


Fig. 1. Overall architecture of HSS-Net.

enhancement but also its simple architecture, remarkably few trainable parameters, and 0.115G FLOPs, which make it both efficient and effective.

B. HSD Module

HSD decomposes the input image into different spectral bands, which capture the physical attributes of the objects in the image [45]. The module uses a convolutional autoencoder (CAE), which learns and reconstructs a low-dimensional representation of the input image. We used the multistage spectral-wise transformer (MST++) from [21] as our decomposition network architecture, which has high efficiency and accuracy, as shown by its RMSE value in the NTIRE 2021 challenge. This network converts an RGB image into 31 spectral bands, creating a richer and more informative spectral representation. The MST++ architecture consists of several single-stage spectral-wise transformers, each with an encoder, a bottleneck, and a decoder. The encoder and the decoder use spectral-wise attention blocks (SABs) and deconvolution layers, respectively. Skip connections link the encoder and decoder layers. We used the pretrained checkpoints from the MST++ network.

C. BS Module

The BS module selects and combines the most relevant and informative bands from the HSI, which reduces the computation time and the redundant information. The module aims to merge the 31 bands into a single grayscale image that focuses on solar panels. The module generates 31 probabilistic weight maps [see Fig. 2(b)] and computes a weighted sum across the bands. The module uses MobileNetV2 [28] as the backbone and DeepLab [11] as the architecture, which are computationally efficient. The module produces 31 outputs with a SoftMax activation function, which *represents* a probability distribution. The module learns and assigns higher weights to the bands that indicate solar panels.

D. Chebyshev Transformation-Based Segmentation Module

In this paragraph, we describe our segmentation module, which is based on CHT layers. We replaced convolutional layers with Chebyshev polynomials, which can reduce the number of trainable parameters and improve the network's performance.

We introduce our segmentation module architecture, which uses EfficientNet [22] as a backbone similar to [26]. U-Net was the best image segmentation network for a long time, but recent works [24], [25] improved it with new pathways and encoders. As the EfficientNet is a computationally efficient backbone, we used it for our segmentation module architecture. We modifed U-Net++ blocks by using depthwise convolutions and scSE blocks (inspired by Jie et al. [27]) to improve computational efficiency and feature focus.

Moreover, we used CHT to reduce the number of trainable parameters and improve segmentation accuracy. CHT applies Chebyshev polynomials to the input, which can handle nonperiodic functions and abrupt changes better than other transformations. For a given signal x, CHT [44] is defined as

$$T(x)_n = \frac{2}{N}\sum_{k=0}^{N-1} x_k T_n\left(\cos\frac{\pi(2k+1)}{2N}\right)$$

where $T(x)_n$ is the nth coefficient of the transformed output, x_k are the elements of the input vector x, and T_n is the nth Chebyshev polynomial. CHT enables a recursive computation of higher order terms, which reduces complexity and improves efficiency. The Chebyshev transformation ensures a well-conditioned basis for function approximation, which minimizes errors and improves robustness. We replaced some convolution layers with CHT layers, which decreases the model's complexity and enhances the segmentation accuracy.

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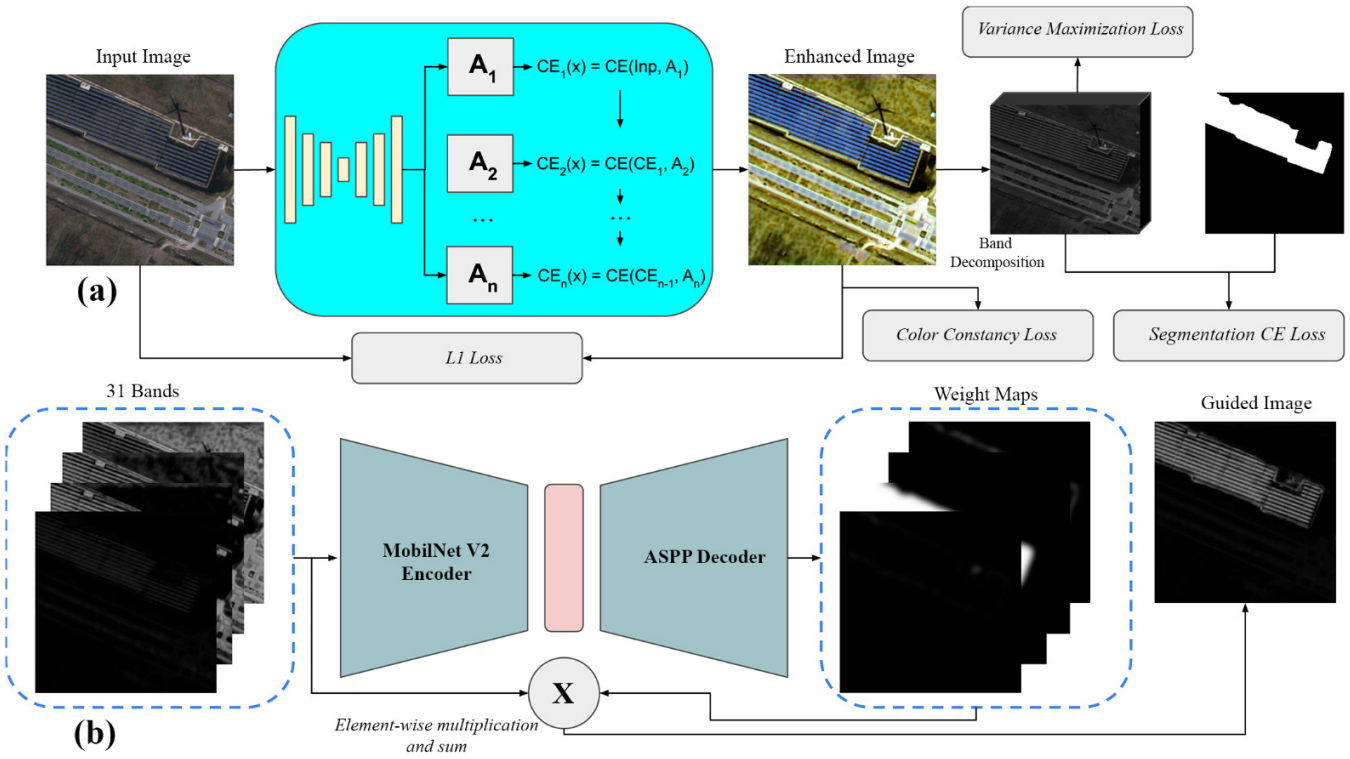


Fig. 2. (a) Unsupervised enhancement module. (b) BS module.

We demonstrate the effectiveness of our method in the experiments (see Section III-D).

E. Loss Functions

Training a network pipeline with separate modules is hard, as each module has its own loss functions and hyperparameters.

Unsupervised Enhancement Module: Enhancing an input image without a corresponding enhanced output label is a complex task. The primary objective is to preserve the semantic information, main colorings, edges, and other details of the image while improving aspects, such as contrast or color. The loss functions used are given in the following.

L1 Loss: To keep the main structure of the image, $L1$ loss compares the original and the output images. This loss keeps the original features and allows some changes.

Variance Maximization Loss: The goal of the enhancement is to spread the information better across the bands after decomposition. This is done by maximizing the variance of the decomposed bands. The decomposition network is fixed, and the loss function updates only the enhancement module. This makes the enhancement module produce an output image that leads to better and more variable decomposition. The formula for variance maximization loss is

$$L_{\text{Var}} = \sum_{i=1}^{H}\sum_{j=1}^{W} \text{Var}_{ij}, \quad \text{Var}_{ij} = \sum_{c=1}^{C} \left(X_{cij} - \mu_{ij}\right)^2$$

where μ_{ij} is the average value across channels for pixel i, j.

Segmentation Cross-Entropy Loss: We enhance the original image, so that at least three bands match the solar panel masks by IoU. We used cross-entropy loss for these bands and masks. The formula is

$$L_{\text{SCE}} = \sum_{i=1}^{3} \text{BCE}(B_i, Y)$$

where BCE is the binary cross entropy loss and B is a tensor with three bands. These bands are selected by top three IoU scores of all bands with label.

Color Constancy Loss: Inspired by Li et al. [23], color constancy loss is crucial to unsupervised enhancement training. It corrects potential color deviations in the enhanced image and maintains the relationship among the three adjusted channels. This loss follows the gray-world color constancy hypothesis [29], if the average color in each sensor channel trends toward gray over the entire image. The formula is given by

$$L_{\text{Col}} = \sum_{p,q\in\varepsilon} \left(J^p - J^q\right)^2, \quad \varepsilon = \{(R,G),(G,B),(R,B)\}.$$

Overall loss function for enhancement module is given by the linear combination of the four components

$$L_{\text{enh}} = \gamma_1 L1 + \gamma_2 L_{\text{var}} + \gamma_3 L_{\text{SCE}} + \gamma_4 L_{\text{col}}.$$

BS and Segmentation Modules: Both the BS and segmentation modules are trained using BCE loss. For the BS module, the objective is to create a guidance image that highlights areas likely to contain solar panels. By employing BCE loss, the module learns to adjust its weight maps to generate an aggregated guidance image that matches the binary label masks as closely as possible. The same loss function is used for the final segmentation module.

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TABLE I
DETAILS OF DATASETS USED FOR TRAINING AND EVALUATION

Dataset	PV01	PV03	PV08	BDAPPV	DeepSolar
Train Set	516	1,820	78	16,779	969
Test Set	122	457	19	4,198	247
Resolution	0.1 m/pixel	0.3 m/pixel	0.8 m/pixel	0.1 m/pixel	-
Image Size	256 x 256	256 x 256	256 x 256	400 x 400	320 x 320

Overall, three main loss functions are used in our training. L_{enh}, $L_{\text{bs_ce}}$, and $L_{\text{segm_ce}}$, where last two are binary cross entropy loss functions.

III. EXPERIMENTS AND RESULTS

A. Datasets

We evaluated on three public SPS datasets (see details in Table I): base de données apprentissage profond PV (BDAPPV) [30]: aerial images from multiple providers with 13 303 400 × 400 0.1-m/pixel Google Earth and 7686 0.2-m/pixel IGN France samples include varying regions and distributions. PV [31]: 3716 satellite/aerial image samples with 0.8-, 0.3-, and 0.1-m/pixel resolutions focused on different PV types and backgrounds. DeepSolar [8]: 1221 satellite images with 320 × 320 resolution over the contiguous US to identify solar PV locations and sizes. Resolution unspecified. These datasets comprise a range of imaging sources, spatial resolutions, geo-distributions, scene types, and PV configurations to support robust and generalized model evaluation. From all datasets, we have removed corrupted images and those images that do not contain any solar panels.

B. Experimental Setup

We used 14 SOTA segmentation methods as baselines, including: 1) three standard architectures (U-Net [32], DeepLabv3+ [11], and PSPNet [12]); 2) six methods for remote sensing images with transformers and attention (dc-Swin [33], MARes-Unet [34], MPRes-Net [35], DCA-Net [36], LA-Net [37], and MA-Net [38]); and 3) five hyperspectral segmentation methods (KDM [39], Adaptor-NAS [40], MSM [41], R-NMF [50], and HAE [51]). The last two methods [50], [51] are used for unmixing hyperspectral data. The first one is based on a nonnegative matrix factorization technique, and the second one is a deep learning-based unmixing technique. After selecting abundance maps, DeepLabv3+ was trained to segment out the final masks. Table II shows some details about the models. We used three metrics to evaluate the segmentation task: kappa coefficient (KI), $F1$ score, and IoU. KI measures the percentage of correctly classified pixels and the chance of agreement. The F1 score measures the harmonic mean of precision and recall. IoU measures the ratio of the intersection and the union of the predicted and ground truth classes. The formula for KI is given as follows:

$$\text{KI} = \frac{2 \times (\text{TP} \times \text{TN} - \text{FN} \times \text{FP})}{(\text{TP} + \text{FP}) \times (\text{FP} + \text{TN}) + (\text{TP} + \text{FN}) \times (\text{FN} + \text{TN})}$$

where TP is the true positive rate, FP is the false positive rate, FN is the false negative rate, and TN is the true negative rate.

TABLE II
DETAILS OF COMPARISON METHODS

Model	Network Type	Encoder	Params
UNet-Resnet34	CNN	Resnet34	24 M
DeepLabV3+	CNN	Resnet50	26 M
PSP-Net	CNN	Resnet50	24 M
DC-Swin	Transformer	Swin	118 M
MARes-Unet	Transformer	Attention Res-Unet	26 M
MPRes-Net	Transformer	Resnet	-
DCA-Net	Transformer	Resnet	69 M
LA-Net	Transformer	Resnet50	23 M
MA-Net	Transformer	Resnet50	35 M
KDM	Hyperspectral	Resnet 18	244 K
Adaptor-NAS	Hyperspectral	Resnet 34	31.9 M
MSM	Hyperspectral	-	-
R-NMF	NMF & CNN	Resnet50	26 M
HAE	DNN & CNN	Resnet50	26 M

Implementation Details: The training of HSS-Net comprises three stages. In the first stage, the enhancement module is trained. After strong computer simulation, the best coefficients of loss functions are chosen as 5, 2, 1, and 3 for γ_1, γ_2, γ_3, and γ_4, respectively. The highest coefficient has the MAE loss component because other loss functions force the output of the enhancement module to have unnatural colors, concentrating more on solar panels and distorting other parts of images, which will degrade the results of the HSD module. The second and third stages are training BS and segmentation modules with a single cross-entropy segmentation loss function. The patch size for the Chebyshev transformation was selected as 16. For all stages, images were either resized to 256 × 256 or randomly cropped to a 256 × 256 patch with a 0.5 probability rate. We also augment the input images by using random image rotation ([0°, 90°, 180°, 270°]), horizontal flip, and vertical flip. For all stages, an Adam optimizer with a learning rate of 0.0005 was used. Epoch counts for enhancement, BS, and segmentation stages were 50, 50, and 150, respectively. For fair comparison, we have trained all 14 models on our datasets while keeping all experimental settings the same as ours, including augmentations, crops, resize, learning rate, and other parameters.

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TABLE III
QUANTITATIVE COMPARISON ON BDAPPV, PV, AND DEEPSOLAR DATASETS

	BDAPPV			**PV**			**DeepSolar**		
	F1	***IoU***	***KI***	***F1***	***IoU***	***KI***	***F1***	***IoU***	***KI***
UNet-Resnet34	0.802	0.720	0.800	0.841	0.794	0.814	0.798	0.715	0.779
DeepLabV3+	0.766	0.677	0.764	0.845	0.796	0.818	0.763	0.684	0.745
PSP-Net	0.788	0.698	0.786	0.833	0.778	0.800	0.776	0.685	0.752
DC-Swin	0.783	0.690	0.781	0.831	0.777	0.802	0.822	0.732	0.803
MARes-Unet	0.861	0.788	0.859	0.878	0.837	0.857	0.914	0.857	0.903
MPRes-Net	0.846	0.769	0.845	0.876	0.831	0.851	0.902	0.841	0.890
DCA-Net	0.726	0.622	0.724	0.830	0.772	0.789	0.846	0.767	0.827
LA-Net	0.834	0.754	0.833	0.846	0.801	0.820	0.884	0.821	0.872
MA-Net	0.862	0.788	0.860	0.877	0.835	0.855	0.916	0.858	0.904
KDM	0.627	0.528	0.625	0.820	0.764	0.788	0.735	0.640	0.713
Adaptor-NAS	0.865	0.791	0.864	0.889	0.845	0.869	0.867	0.809	0.855
MSM	0.763	0.658	0.731	0.470	0.399	0.310	0.753	0.649	0.724
R-NMF	0.788	0.698	0.791	0.869	0.835	0.846	0.868	0.800	0.853
HAE	0.792	0.716	0.823	0.871	0.827	0.855	0.892	0.797	0.884
HSS-Net	**0.883**	**0.816**	**0.881**	**0.893**	**0.853**	**0.871**	**0.922**	**0.868**	**0.912**

TABLE IV
QUANTITATIVE COMPARISONS ON PV SUBSETS

	PV01			**PV03**			**PV08**		
	F1	**IoU**	**KI**	**F1**	**IoU**	**KI**	**F1**	**IoU**	**KI**
UNet-Resnet34	0.889	0.841	0.858	0.857	0.810	0.831	0.158	0.118	0.146
DeepLabV3+	0.874	0.819	0.842	0.866	0.819	0.839	0.174	0.129	0.167
PSP-Net	0.871	0.807	0.828	0.847	0.794	0.816	0.283	0.208	0.264
DC-Swin	0.870	0.819	0.837	0.839	0.787	0.811	0.394	0.294	0.374
MARes-Unet	0.904	0.870	0.884	0.887	0.845	0.864	0.530	0.441	0.519
MPRes-Net	0.884	0.842	0.854	0.886	0.842	0.862	0.581	0.483	0.567
DCA-Net	0.840	0.766	0.771	0.844	0.793	0.810	0.432	0.328	0.416
LA-Net	0.888	0.849	0.862	0.854	0.808	0.827	0.411	0.339	0.399
MA-Net	0.905	0.868	0.883	0.882	0.841	0.859	0.595	0.498	0.578
KDM	0.839	0.774	0.794	0.842	0.789	0.812	0.206	0.139	0.189
Adaptor-NAS	0.910	0.882	0.891	0.899	0.857	0.879	0.443	0.335	0.427
MSM	0.471	0.387	0.262	0.485	0.416	0.334	0.121	0.078	0.060
R-NMF	0.887	0.844	0.863	0.853	0.800	0.814	0.482	0.371	0.375
HAE	0.894	0.852	0.871	0.848	0.782	0.804	0.493	0.381	0.384
HSS-Net	**0.916**	**0.882**	**0.895**	**0.9**	**0.860**	**0.881**	**0.6**	**0.504**	**0.587**

C. Comparison With Other Methods

We compared HSS-Net with 14 segmentation models, including convolution-based, transformer-based, and HSI segmentation approaches. For fair comparison, we used the same experimental settings and our decomposition of bands for all methods. We used three binary segmentation metrics and three datasets. Tables III and IV summarize the results of the quantitative comparison on three datasets. HSS-Net outperformed all other SOTA models on three datasets (and on the subsets of the PV dataset in Table IV) on three different segmentation measures. Due to the limited size of the PV08 dataset (19 images), we acknowledge the possibility of statistical anomalies for that test set in our evaluation metrics. Therefore, we supplement our quantitative assessment with a qualitative analysis, visually showing the superior performance of HSS-Net, including on small-size panels from PV08 (see the third image of Fig. 3). We show the predicted masks of our method and other methods, as well as ground truth masks (Fig. 3). Three images are selected from the DeepSolar, BDAPPV, and PV datasets from top to bottom. For the first two images, convolutional neural network (CNN)-based methods (DeepLabv3+, UNet, and PSPNet) identified general areas with blurred masks with numerous false negative pixels. In contrast, transformer-based approaches yielded less blurred regions in their masks, but they were prone to a higher rate of false positives (especially for the second image). They predicted more distinct shapes of positive areas compared with CNN-based approaches, but they still lack precise boundaries. Hyperspectral-based

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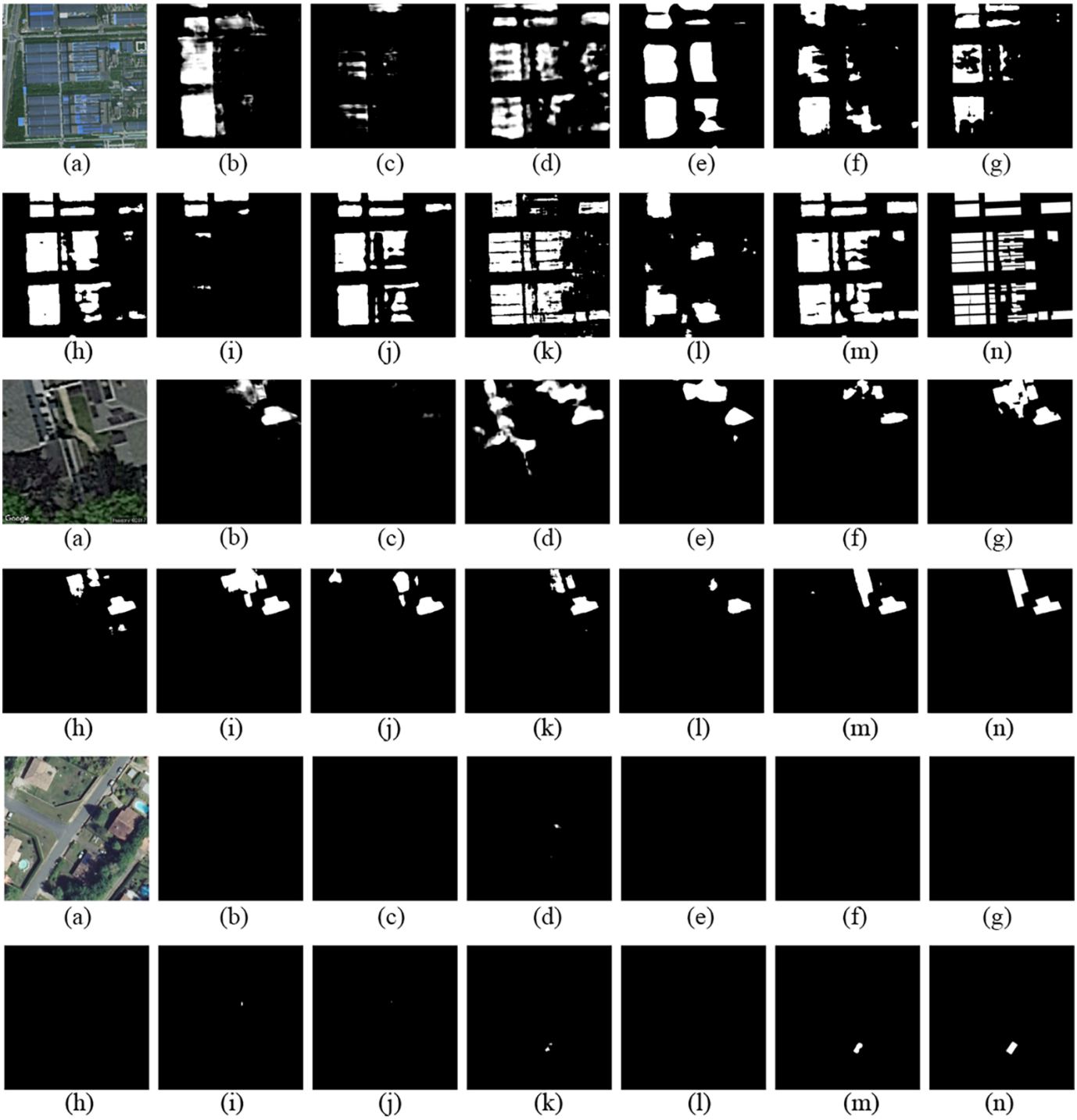


Fig. 3. Qualitative comparison of the proposed network with SOTA networks. (a) Input image. (b) UNet. (c) DeepLabV3+. (d) PSPNet. (e) DCA-Net. (f) DC-Swin, (g) LA-Net. (h) MA-Net. (i) MARes-UNet. (j) MPResNet. (k) AdaptorNAS. (l) KDM. (m) HSS-Net. (n) ground truth.

segmentation methods succeeded in predicting better masks compared with standard segmentation models, as expected. For instance, AdaptorNAS has managed to find most of the solar panel regions on the first and second images, but it still exhibits false negatives in certain local areas, where solar panel coverage is underestimated due to its architecture's loss of global semantic context. On the other hand, KDM predicted less local false positives, but missed some broader areas of panels. Our predicted masks have closely mirrored ground truth masks, capturing all crucial regions without missing any important parts. They have also successfully minimized false positives, demonstrating an accurate segmentation capability that outperforms other methods.

D. Ablation Study

In this section, we examine the impact of each component of the proposed method. Table V summarizes the results across

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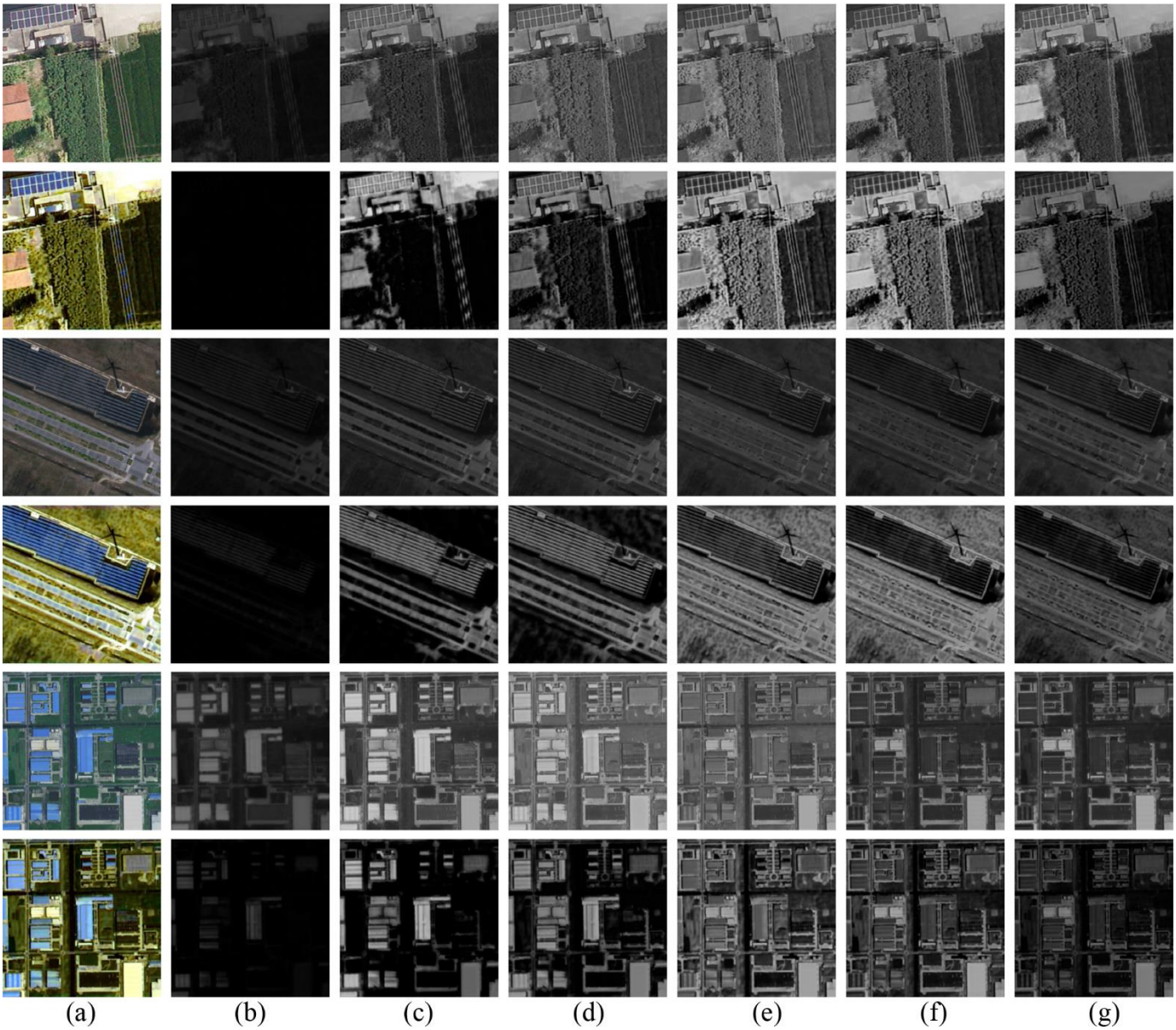


Fig. 4. Band decomposition results for some band indexes. (a) Input and enhanced images. (b) Index 1. (c) Index 5. (d) Index 10. (e) Index 15. (f) Index 25. (g) Index 30.

three datasets by progressively eliminating the enhancement, decomposition, and BS blocks. It can be observed from the numbers in Table V that each block has its own significance.

1) HSD: Fig. 4 illustrates the original image with its enhanced version (a) along with some of the bands (1, 5, 10, 15, 25, and 30 out of a total of 31 bands). Each band has its own unique characteristics and features, particularly for solar panels. The fifth band highlights mostly solar panels, containing panel-related information. The tenth and 15th bands reveal low-level fine details of features, and the 15th and 25th bands focus more on green elements like grass. This is the proof of concept for our HSD approach, demonstrating how different bands capture distinct aspects of information and allowing us to select specific bands for our task later.

2) Enhancement Module: The effectiveness of the decomposition mostly depends on the quality of the initial enhancement stage. Fig. 4(c)–(g) presents the impact of our enhancement module. Without enhancement, the bands lack distinct information and appear similar. The enhancement's role is crucial in highlighting the colors and ensuring successful decomposition later.

3) BS Module: The band-selection module generates weighted maps for each of the 31 bands. Some are displayed in Fig. 5(j)–(n) and their corresponding bands in Fig. 5(c)–(g). It can be observed that bands with higher concentrations of solar panel data [Fig. 5(d)] have weight maps highlighting those areas. Similarly, bands that are more effective in showing edge features and boundaries [Fig. 5(e) and (f)] have their corresponding weight maps highlighting those boundary regions [Fig. 5(l) and (m)]. The final guide band [Fig. 5(i)] is formulated through a weighted average of the bands with their corresponding weight maps. The final segmentation output (h)

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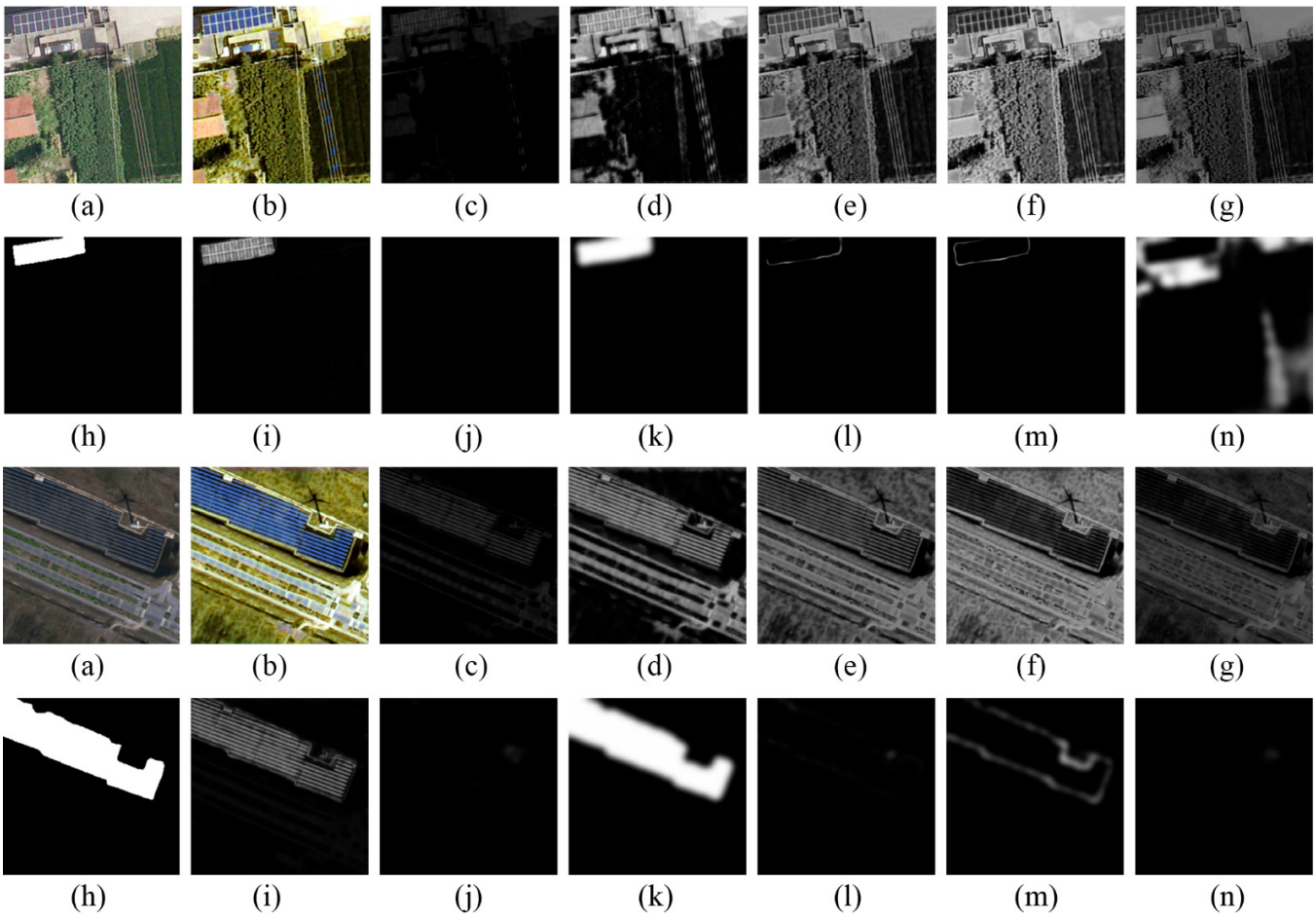


Fig. 5. (c)–(g) Band decomposition with their (j)–(n) corresponding weight maps, predicted by band selection module. (a) and (b) Original image and its enhanced version, respectively. (i) Combined guided image. (h) Final mask.

TABLE V
QUANTITATIVE RESULTS OF ABLATION STUDY

Dataset	**PV**		**BDAPPV**		**DeepSolar**	
Metric	***F1***	***IoU***	***F1***	***IoU***	***F1***	***IoU***
W/O Enhancement	0.767	0.716	0.854	0.768	0.905	0.831
W/O Hyp. Decomposition	0.789	0.726	0.863	0.784	0.911	0.848
W/O Band Selection	0.784	0.714	0.874	0.764	0.895	0.841
Proposed Method	**0.893**	**0.853**	**0.883**	**0.816**	**0.922**	**0.868**

is obtained by feeding the enhanced input image (b) and the constructed guide band into the segmentation network.

4) Chebyshev Transformation: Utilizing Chebyshev transformation in our segmentation network significantly reduces the number of trainable parameters. Remarkably, such modifications maintain segmentation accuracy while drastically decreasing the parameter count (from **11** to **3.5 M**). A smaller number of trainable parameters shortens the training time and prevents the model from overfitting. Furthermore, this approach enhances the model's efficiency, making it more suitable for deployment in resource-limited environments.

IV. CONCLUSION

We proposed a novel framework for SPS from remote sensing images. Our method uses four modules: HSD, unsupervised enhancement, BS, and segmentation module based on CHT. We tested it on three SPS datasets and compared it with SOTA methods, using IoU, $F1$-score, and KI as metrics. We also analyzed the effect of each module in our method. Our method: 1) outperformed SOTA methods in accuracy, efficiency, and stability; 2) can improve segmentation in remote sensing other applications by reducing computational load and enhancing image decomposition; and 3) provides a practical and scalable solution for SPS, which opens new opportunities and challenges for future research, such as exploring HSD and applying our method to other segmentation tasks. We will study PV panel segmentation with rooftop solar PV potential assessments and installation suggestions in the future. We will also explore synthetic HSI decomposition and optimize deep learning techniques for efficient band selection utilizing hyperspectral unmixing techniques.

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REFERENCES

[1] M. R. S. Shaikh, "A review paper on electricity generation from solar energy," *Int. J. Res. Appl. Sci. Eng. Technol.*, no. 9, pp. 1884–1889, Sep. 2017, doi: 10.22214/ijraset.2017.9272.

[2] J. Wang et al., "Rooftop PV segmenter: A size-aware network for segmenting rooftop photovoltaic systems from high-resolution imagery," *Remote Sens.*, vol. 15, no. 21, p. 5232, Nov. 2023, doi: 10.3390/rs15215232.

[3] H. Tan et al., "Enhancing PV panel segmentation in remote sensing images with constraint refinement modules," *Appl. Energy*, vol. 350, Nov. 2023, Art. no. 121757, doi: 10.1016/j.apenergy.2023.121757.

[4] M. V. C. V. D. Costa et al., "Remote sensing for monitoring photovoltaic solar plants in Brazil using deep semantic segmentation," *Energies*, vol. 14, no. 10, p. 2960, May 2021, doi: 10.3390/en14102960.

[5] J. Zhang, X. Jia, and J. Hu, "SP-RAN: Self-paced residual aggregated network for solar panel mapping in weakly labeled aerial images," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5612715, doi: 10.1109/TGRS.2021.3123268.

[6] L. Li and E. Lau, "RU-Net: Solar panel detection from remote sensing image," in *Proc. IEEE Green Energy Smart Syst. Syst. (IGESSC)*, Nov. 2022, pp. 1–6, doi: 10.1109/IGESSC55810.2022.9955325.

[7] J. M. Malof, K. Bradbury, L. M. Collins, and R. G. Newell, "Automatic detection of solar photovoltaic arrays in high resolution aerial imagery," *Appl. Energy*, vol. 183, pp. 229–240, Dec. 2016, doi: 10.1016/j.apenergy.2016.08.191.

[8] J. Yu, Z. Wang, A. Majumdar, and R. Rajagopal, "DeepSolar: A machine learning framework to efficiently construct a solar deployment database in the United States," *Joule*, vol. 2, no. 12, pp. 2605–2617, Dec. 2018, doi: 10.1016/j.joule.2018.11.021.

[9] J. Long, E. Shelhamer, and T. Darrell, "Fully convolutional networks for semantic segmentation," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Boston, MA, USA, Jun. 2015, pp. 3431–3440.

[10] H. Ye, S. Liu, K. Jin, and H. Cheng, "CT-UNet: An improved neural network based on U-Net for building segmentation in remote sensing images," in *Proc. 25th Int. Conf. Pattern Recognit. (ICPR)*, Milan, Italy, Jan. 2021, pp. 166–172.

[11] L.-C. Chen, Y. Zhu, G. Papandreou, F. Schroff, and H. Adam, "Encoder-decoder with atrous separable convolution for semantic image segmentation," in *Computer Vision—ECCV* (Lecture Notes in Computer Science), vol. 11211, V. Ferrari, M. Hebert, C. Sminchisescu, and Y. Weiss, Eds. Cham, Switzerland: Springer, 2018, pp. 833–851.

[12] H. Zhao, J. Shi, X. Qi, X. Wang, and J. Jia, "Pyramid scene parsing network," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jul. 2017, pp. 6230–6239.

[13] L. Zhuang, Z. Zhang, and L. Wang, "The automatic segmentation of residential solar panels based on satellite images: A cross learning driven U-Net method," *Appl. Soft Comput.*, vol. 92, Jul. 2020, Art. no. 106283, doi: 10.1016/j.asoc.2020.106283.

[14] J. Yuan, H. L. Yang, O. A. Omitaomu, and B. L. Bhaduri, "Large-scale solar panel mapping from aerial images using deep convolutional networks," in *Proc. IEEE Int. Conf. Big Data (Big Data)*, Washington, DC, USA, Dec. 2016, pp. 2703–2708.

[15] V. Golovko, S. Bezobrazov, A. Kroshchanka, A. Sachenko, M. Komar, and A. Karachka, "Convolutional neural network based solar photovoltaic panel detection in satellite photos," in *Proc. 9th IEEE Int. Conf. Intell. Data Acquisition Adv. Comput. Syst. Technol. Appl. (IDAACS)*, vol. 1, Bucharest, Romania, Sep. 2017, pp. 14–19.

[16] A. de Luis et al., "SolarFormer: Multi-scale transformer for solar PV profiling," 2023, *arXiv:2310.20057*.

[17] Z. Guo et al., "TransPV: Refining photovoltaic panel detection accuracy through a vision transformer-based deep learning model," *Appl. Energy*, vol. 355, Feb. 2024, Art. no. 122282, doi: 10.1016/j.apenergy.2023.122282.

[18] C. Jörges, H. S. Vidal, T. Hank, and H. Bach, "Detection of solar photovoltaic power plants using satellite and airborne hyperspectral imaging," *Remote Sens.*, vol. 15, no. 13, p. 3403, Jul. 2023, doi: 10.3390/rs15133403.

[19] D. Czirjak, "Detecting photovoltaic solar panels using hyperspectral imagery and estimating solar power production," *J. Appl. Remote Sens.*, vol. 11, no. 2, Apr. 2017, Art. no. 026007, doi: 10.1117/1.jrs.11.026007.

[20] B. Arad et al., "NTIRE 2022 spectral recovery challenge and data set," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. Workshops (CVPRW)*, New Orleans, LA, USA, Jun. 2022, pp. 862–880, doi: 10.1109/CVPRW56347.2022.00102.

[21] Y. Cai et al., "MST++: Multi-stage spectral-wise transformer for efficient spectral reconstruction," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. Workshops (CVPRW)*, Jun. 2022, pp. 744–754.

[22] M. Tan and Q. V. Le, "EfficientNet: Rethinking model scaling for convolutional neural networks," 2019, *arXiv:1905.11946*.

[23] C. Li, C. Guo, and C. C. Loy, "Learning to enhance low-light image via zero-reference deep curve estimation," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 44, no. 8, pp. 4225–4238, Aug. 2022, doi: 10.1109/TPAMI.2021.3063604.

[24] Z. Zhou, M. M. R. Siddiquee, N. Tajbakhsh, and J. Liang, "UNet++: Redesigning skip connections to exploit multiscale features in image segmentation," *IEEE Trans. Med. Imag.*, vol. 39, no. 6, pp. 1856–1867, Jun. 2020, doi: 10.1109/TMI.2019.2959609.

[25] S. Yang, J. Kweon, and Y.-H. Kim, "Major vessel segmentation on X-ray coronary angiography using deep networks with a novel penalty loss function," in *Proc. MIDL*, Apr. 2019, pp. 1–5.

[26] J. L. Silva, M. N. Menezes, T. Rodrigues, B. Silva, F. J. Pinto, and A. L. Oliveira, "Encoder–decoder architectures for clinically relevant coronary artery segmentation," 2021, *arXiv:2106.11447*.

[27] J. Hu, L. Shen, and G. Sun, "Squeeze-and-excitation networks," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, Jun. 2018, pp. 7132–7141, doi: 10.1109/CVPR.2018.00745.

[28] M. Sandler, A. Howard, M. Zhu, A. Zhmoginov, and L.-C. Chen, "MobileNetV2: Inverted residuals and linear bottlenecks," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, Jun. 2018, pp. 4510–4520, doi: 10.1109/CVPR.2018.00474.

[29] G. Buchsbaum, "A spatial processor model for object colour perception," *J. Franklin Inst.*, vol. 310, no. 1, pp. 1–26, Jul. 1980, doi: 10.1016/0016-0032(80)90058-7.

[30] G. Kasmi et al., "A crowdsourced dataset of aerial images with annotated solar photovoltaic arrays and installation metadata," *Sci. Data*, vol. 10, no. 1, p. 59, Jan. 2023.

[31] H. Jiang et al., "Multi-resolution dataset for photovoltaic panel segmentation from satellite and aerial imagery," *Earth Syst. Sci. Data*, vol. 13, no. 11, pp. 5389–5401, Nov. 2021.

[32] N. Siddique, S. Paheding, C. P. Elkin, and V. Devabhaktuni, "U-Net and its variants for medical image segmentation: A review of theory and applications," *IEEE Access*, vol. 9, pp. 82031–82057, 2021, doi: 10.1109/ACCESS.2021.3086020.

[33] L. Wang, R. Li, C. Duan, C. Zhang, X. Meng, and S. Fang, "A novel transformer based semantic segmentation scheme for fine-resolution remote sensing images," *IEEE Geosci. Remote Sens. Lett.*, vol. 19, pp. 1–5, 2022, doi: 10.1109/LGRS.2022.3143368.

[34] R. Li, S. Zheng, C. Duan, J. Su, and C. Zhang, "Multistage attention ResU-net for semantic segmentation of fine-resolution remote sensing images," *IEEE Geosci. Remote Sens. Lett.*, vol. 19, pp. 1–5, Mar. 2022, doi: 10.1109/LGRS.2021.3063381.

[35] L. Ding et al., "MP-ResNet: Multipath residual network for the semantic segmentation of high-resolution PolSAR images," *IEEE Geosci. Remote Sens. Lett.*, vol. 19, pp. 1–5, 2022, doi: 10.1109/LGRS.2021.3079925.

[36] L. Wu, M. Lu, and L. Fang, "Deep covariance alignment for domain adaptive remote sensing image segmentation," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5620811, doi: 10.1109/TGRS.2022.3163278.

[37] L. Ding, H. Tang, and L. Bruzzone, "LANet: Local attention embedding to improve the semantic segmentation of remote sensing images," *IEEE Trans. Geosci. Remote Sens.*, vol. 59, no. 1, pp. 426–435, Jan. 2021, doi: 10.1109/TGRS.2020.2994150.

[38] R. Li et al., "Multiattention network for semantic segmentation of fine-resolution remote sensing images," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5607713, doi: 10.1109/TGRS.2021.3093977.

[39] M. H. Phan, S. L. Phung, K. Luu, and A. Bouzerdoum, "Efficient hyperspectral image segmentation for biosecurity scanning using knowledge distillation from multi-head teacher," *Neurocomputing*, vol. 504, pp. 189–203, Sep. 2022, doi: 10.1016/j.neucom.2022.06.095.

[40] S. P. Ang, S. L. Phung, L. Bui, and A. Bouzerdoum, "AdaptorNAS: A new perturbation-based neural architecture search for hyperspectral image segmentation," *IEEE Trans. Circuits Syst. Video Technol.*, vol. 34, no. 3, pp. 1559–1571, Mar. 2024, doi: 10.1109/TCSVT.2023.3298796.

[41] J.-C. Cohrs, C. Bajaj, and B. Berkels, "A distribution-dependent Mumford–Shah model for unsupervised hyperspectral image segmentation," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5545121, doi: 10.1109/TGRS.2022.3227061.

[42] C. Pan, "Gibbs phenomenon removal and digital filtering directly through the fast Fourier transform," *IEEE Trans. Signal Process.*, vol. 49, no. 2, pp. 444–448, Feb. 2001, doi: 10.1109/78.902128.

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[43] T. Trongtirakul and S. Agaian, "Unsupervised and optimized thermal image quality enhancement and visual surveillance applications," *Signal Process., Image Commun.*, vol. 105, Jul. 2022, Art. no. 116714, doi: 10.1016/j.image.2022.116714.

[44] T. J. Rivlin, *Chebyshev Polynomials*. New York, NY, USA: Courier Dover, 2020.

[45] L. Liu, W. Li, Z. Shi, and Z. Zou, "Physics-informed hyperspectral remote sensing image synthesis with deep conditional generative adversarial networks," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5528215, doi: 10.1109/TGRS.2022.3173532.

[46] R. Zhu et al., "Deep solar PV refiner: A detail-oriented deep learning network for refined segmentation of photovoltaic areas from satellite imagery," *Int. J. Appl. Earth Observ. Geoinf.*, vol. 116, Feb. 2023, Art. no. 103134.

[47] F. Li, W. Dong, and W. Wu, "A general model for comprehensive electrical characterization of photovoltaics under partial shaded conditions," *Adv. Appl. Energy*, vol. 9, Feb. 2023, Art. no. 100118.

[48] M. Schulz, B. Boughattas, and F. Wendel, "DetEEktor: Mask R-CNN based neural network for energy plant identification on aerial photographs," *Energy AI*, vol. 5, Sep. 2021, Art. no. 100069.

[49] Z. Guo et al., "Accurate and generalizable photovoltaic panel segmentation using deep learning for imbalanced datasets," *Renew. Energy*, vol. 219, Dec. 2023, Art. no. 119471.

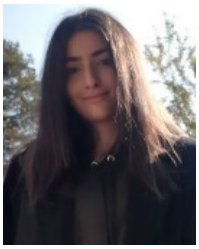
[50] C. Févotte and N. Dobigeon, "Nonlinear hyperspectral unmixing with robust nonnegative matrix factorization," *IEEE Trans. Image Process.*, vol. 24, no. 12, pp. 4810–4819, Dec. 2015, doi: 10.1109/TIP.2015.2468177.

[51] B. Palsson, J. Sigurdsson, J. R. Sveinsson, and M. O. Ulfarsson, "Hyperspectral unmixing using a neural network autoencoder," *IEEE Access*, vol. 6, pp. 25646–25656, 2018, doi: 10.1109/ACCESS.2018.2818280.



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